

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 3 - Visualization Design Activity
Weightage:	35%	Total Marks:	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
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Section / Batch:	24 Batch / AI&DS	Date of Submission:	28-08-2026
Faculty In-charge:	K.Veena devi	Assessment Mode:	Individual Submission

Code of Conduct

I ____U.Ananya(192424382)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: ____U.Ananya_____

Activity 1 – Basic Plot Creation

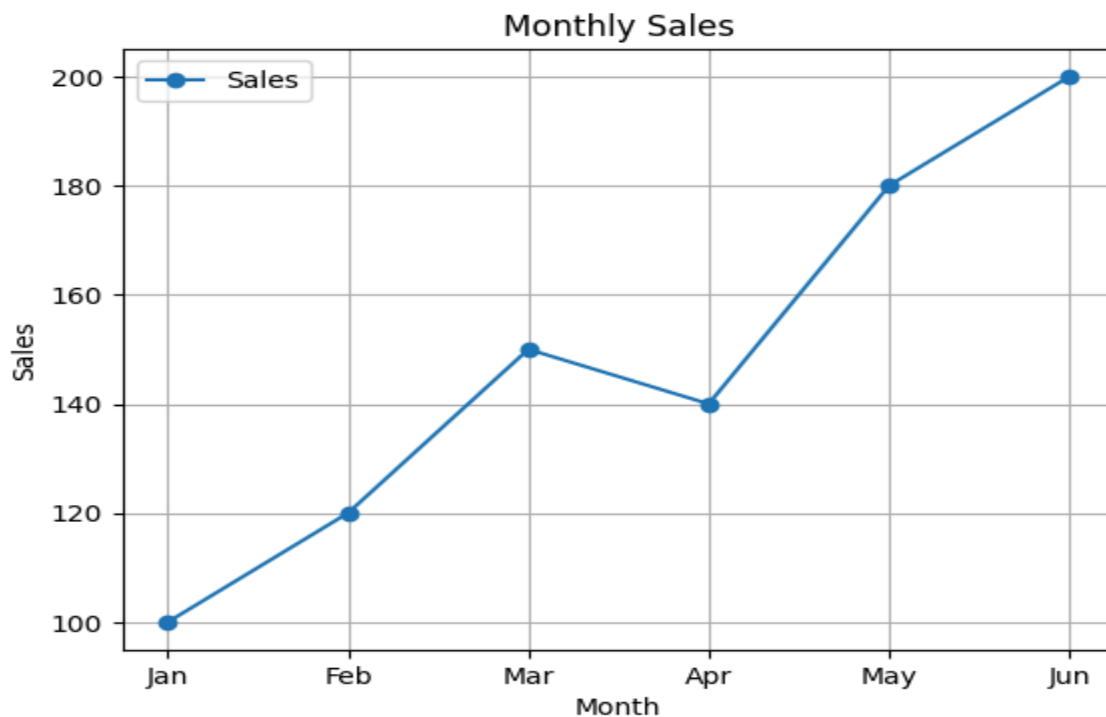
Aim

Create a line plot showing monthly sales.

Code

```
import matplotlib.pyplot as plt  
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]  
sales = [100, 120, 150, 140, 180, 200]  
plt.plot(months, sales, marker="o", label="Sales")  
plt.title("Monthly Sales")  
plt.xlabel("Month")  
plt.ylabel("Sales")  
plt.legend()  
plt.grid()  
plt.show()
```

Output



Conclusion

Sales generally show an increasing trend, although there is a small decrease in April.

Activity 3 – Compare Visualization Types

Aim

Use the same data to create a line plot, bar plot, and scatter plot.

Code

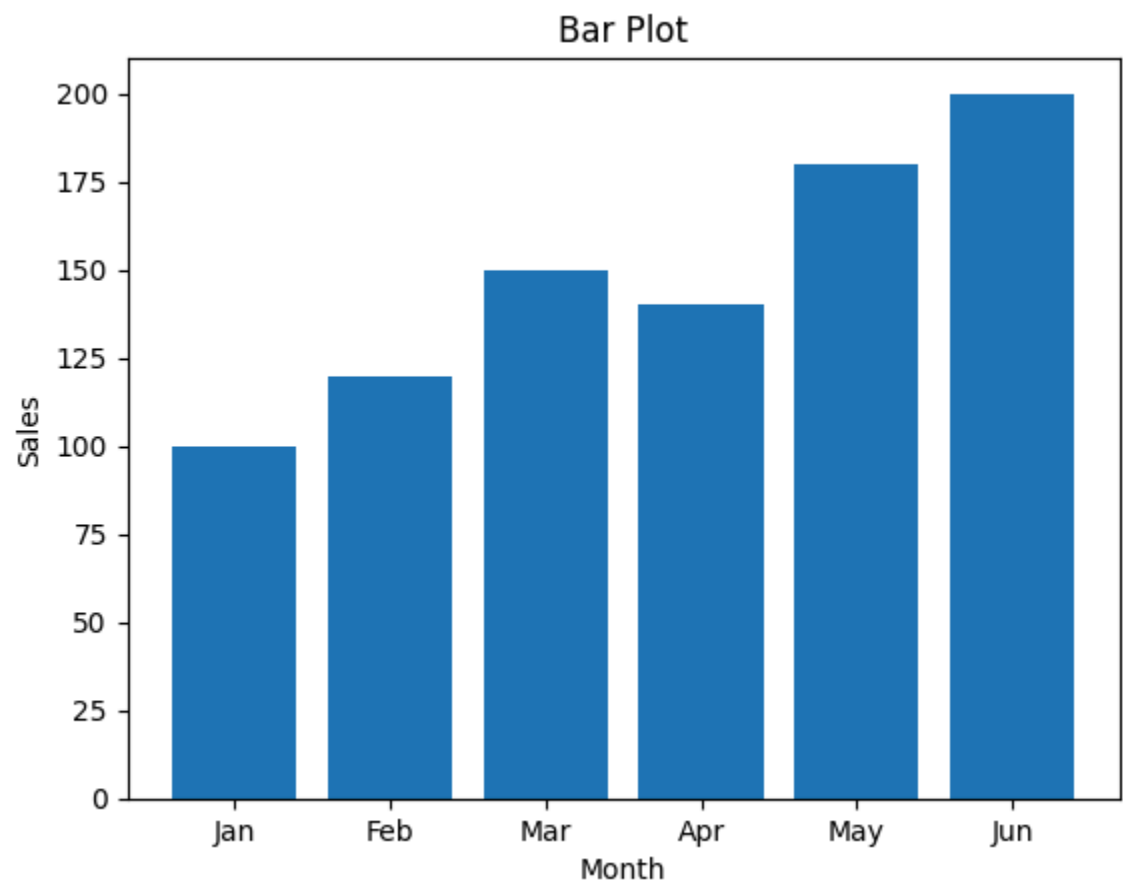
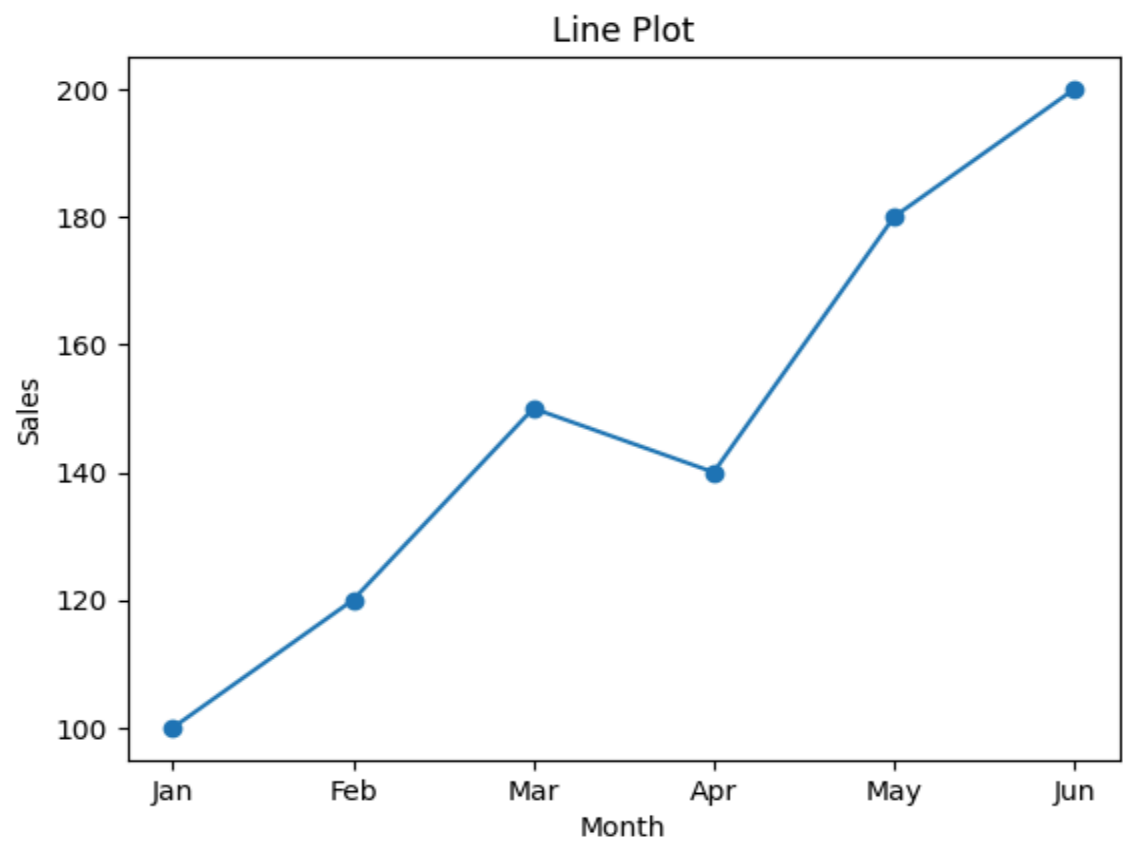
```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [100, 120, 150, 140, 180, 200]

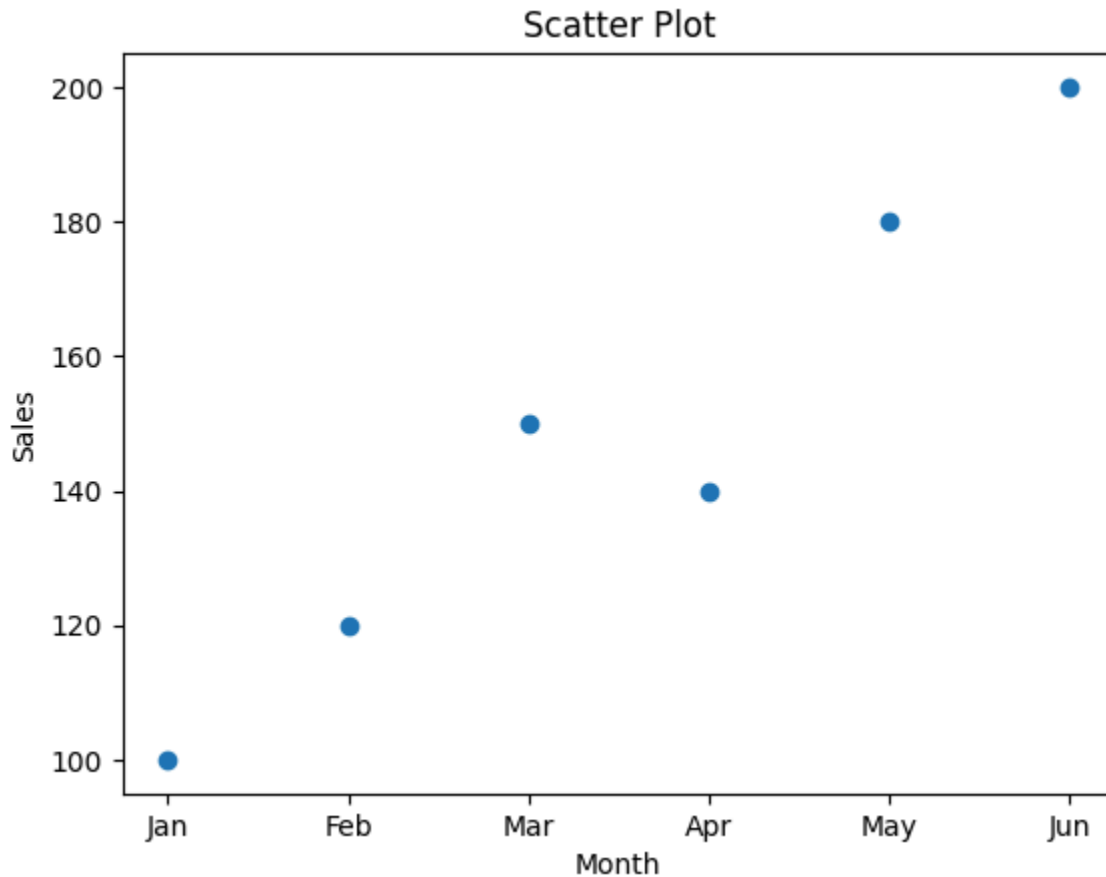
# Line plot
plt.plot(months, sales, marker="o")
plt.title("Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

# Bar plot
plt.bar(months, sales)
plt.title("Bar Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

# Scatter plot
plt.scatter(months, sales)
plt.title("Scatter Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Output:





Justification

- Line plot → Best for showing sales trends over time.
- Bar plot → Best for comparing sales between months.
- Scatter plot → Best for showing individual data points.

Conclusion

For monthly sales data, the line plot is most effective because it clearly shows the change in sales over time.

Activity 4 – Pandas Visualization

Aim

Create a DataFrame containing Product, Sales, and Profit and visualize the data using Pandas.

Code

```
import pandas as pd
```

```
import matplotlib.pyplot as plt

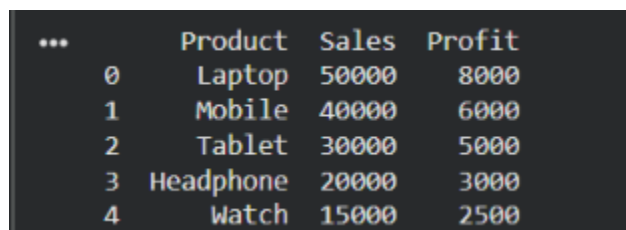
df = pd.DataFrame({
    "Product": ["Laptop", "Mobile", "Tablet", "Headphone", "Watch"],
    "Sales": [50000, 40000, 30000, 20000, 15000],
    "Profit": [8000, 6000, 5000, 3000, 2500]
})

print(df)

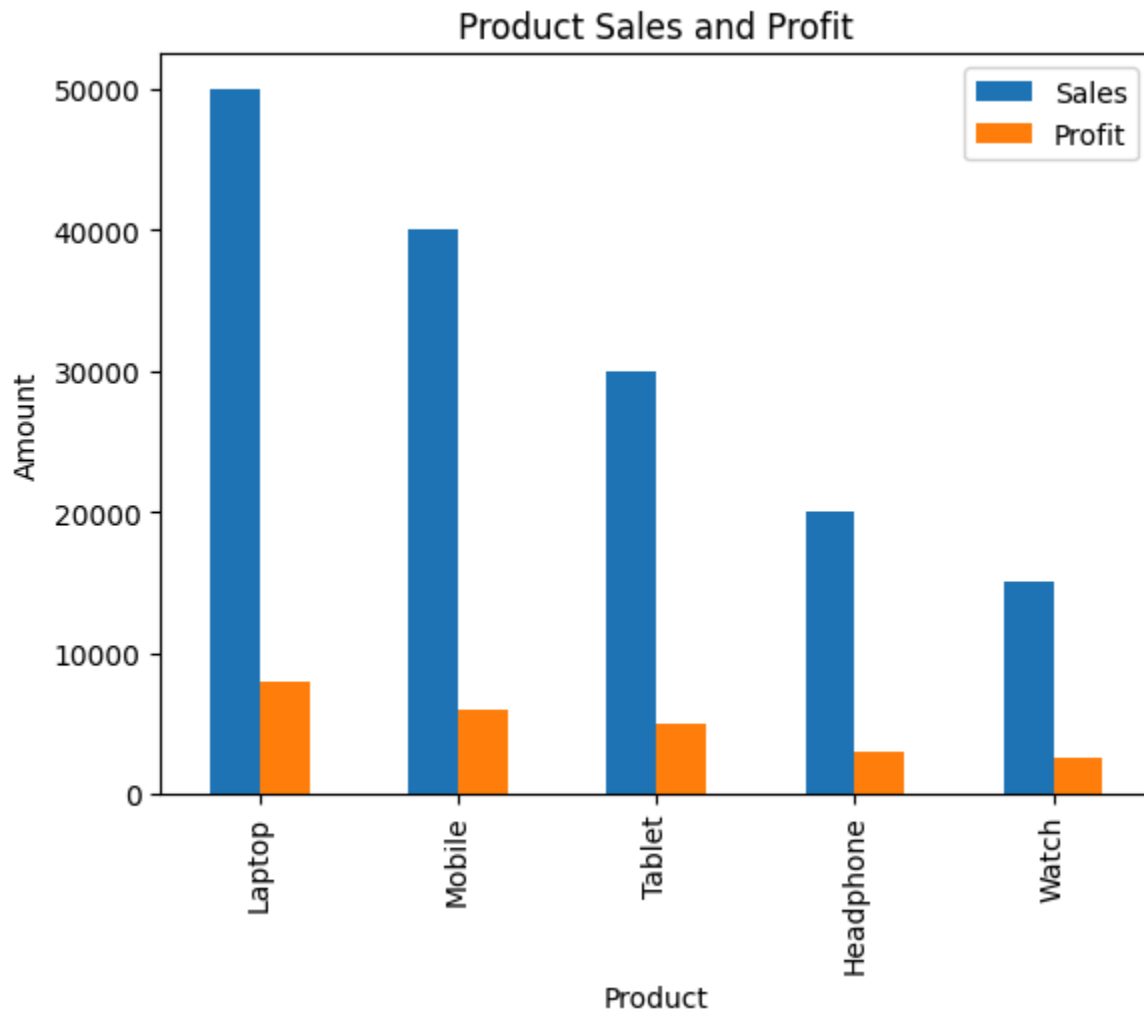
df.plot(
    x="Product",
    y=["Sales", "Profit"],
    kind="bar"
)

plt.title("Product Sales and Profit")
plt.xlabel("Product")
plt.ylabel("Amount")
plt.show()
```

Output:



...	Product	Sales	Profit
0	Laptop	50000	8000
1	Mobile	40000	6000
2	Tablet	30000	5000
3	Headphone	20000	3000
4	Watch	15000	2500



Conclusion

The graph compares sales and profit for different products. The Laptop has the highest sales and profit among the products.

Activity 6 – Relationship Visualization

Aim

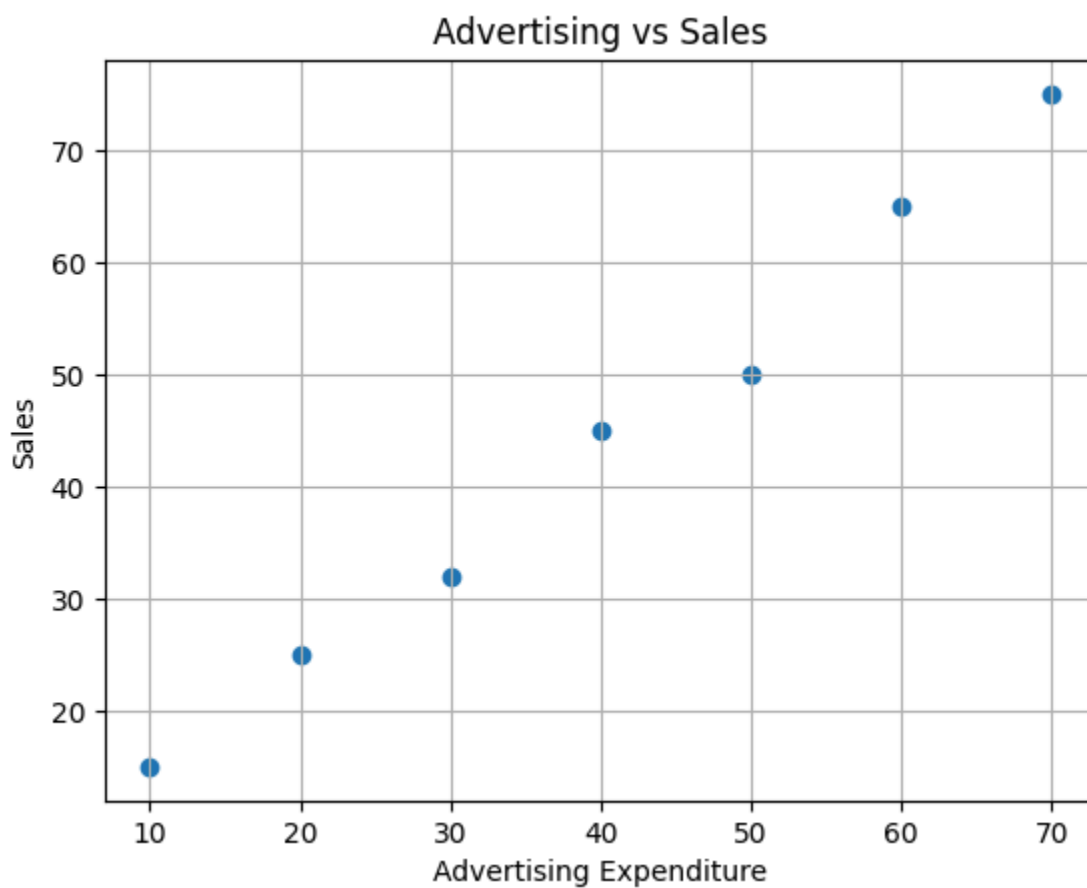
Analyze the relationship between advertising expenditure and sales using a scatter plot.

Code

```
import matplotlib.pyplot as plt
advertising = [10, 20, 30, 40, 50, 60, 70]
sales = [15, 25, 32, 45, 50, 65, 75]
```

```
plt.scatter(advertising, sales)
plt.title("Advertising vs Sales")
plt.xlabel("Advertising Expenditure")
plt.ylabel("Sales")
plt.grid()
plt.show()
```

Output:



Interpretation

The points generally move upward from left to right.

This indicates a positive relationship between advertising expenditure and sales.

Conclusion

As advertising expenditure increases, sales generally increase. Therefore, the dataset shows a positive correlation.

Activity 7 – Distribution Visualization

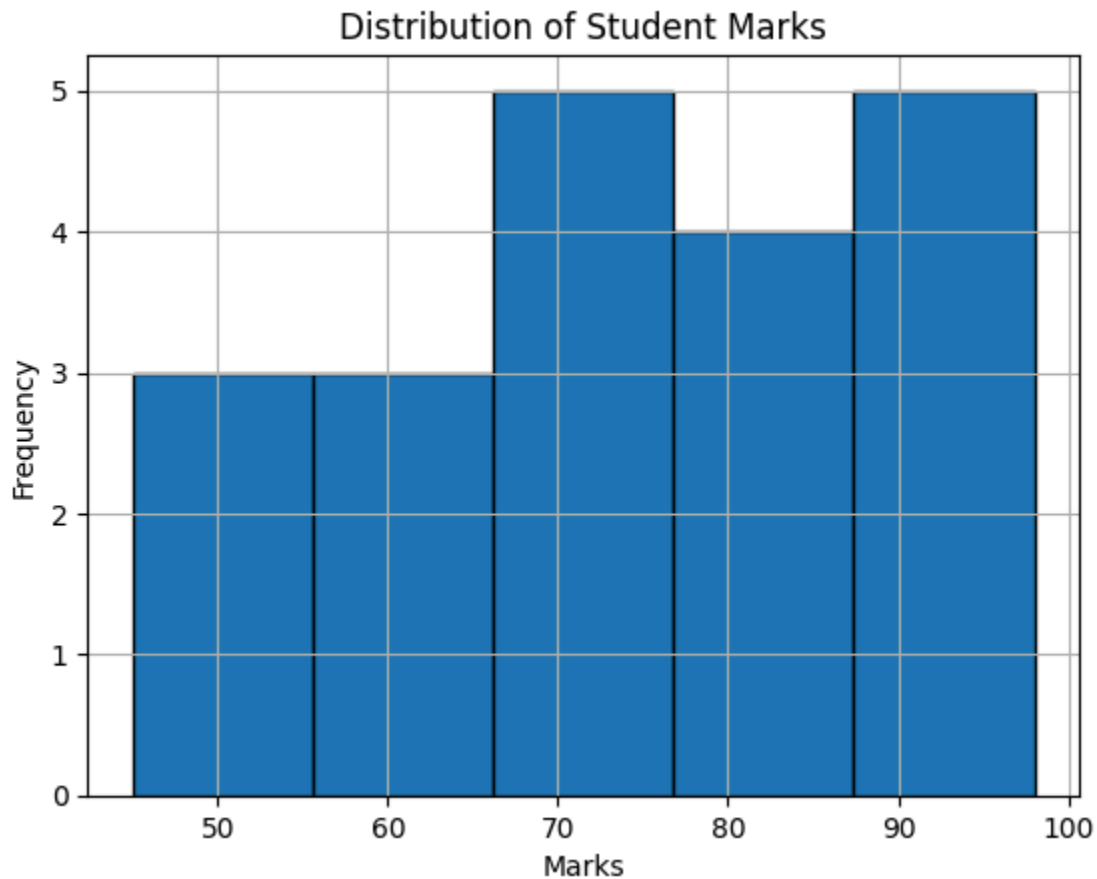
Aim

Create a histogram of student marks and analyze the distribution.

Code

```
import matplotlib.pyplot as plt
marks = [
    45, 50, 55, 60, 62,
    65, 68, 70, 72, 75,
    76, 78, 80, 82, 85,
    88, 90, 92, 95, 98
]
plt.hist(marks, bins=5, edgecolor="black")
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.grid()
plt.show()
```

Output:



Observations

- Most students scored between 60 and 90.
- The center of the distribution is approximately around 70–80 marks.
- The spread is from approximately 45 to 98.
- There are no major extreme outliers.
- The distribution is approximately balanced, with more students concentrated around the middle and higher marks.

Conclusion

The histogram provides a simple view of the distribution, center, and spread of student marks.